

Logan Luna

727-291-2456
Logan03Luna@gmail.com
DoD SMART Scholar

LinkedIn: www.linkedin.com/in/logan-luna
GitHub: <https://github.com/Logan-Cole-Luna>
Active Security Clearance Level: Secret

Education

Embry-Riddle Aeronautical University | Honors Program | Dean's List **Expected Graduation: May 2026**

- ❖ **Degree:** Bachelor of Science in Computer Science
- ❖ **Minor:** Computational Mathematics
- ❖ **GPA:** 3.8

Professional Experience

Intern

Department of Defense, Beam and Optic Control for High Energy Weapons **2024–Present**
Naval Surface Warfare Center Dahlgren Division (NSWCDD)

- ❖ Developed a real-time autonomous object tracking system using YOLO (object detection), ORB (keypoint matching), optical flow via homography estimation, reidentification (reID), and Kalman filtering
- ❖ Created custom neural network architecture to handle temporal information, enhancing operational efficiency
- ❖ Integrated ML, CV, and radar data to improve detection precision and tracking reliability in dynamic environments

Researcher

Satellite Trajectory Optimization Research **2024–Present**

- ❖ Applied reinforcement learning to optimize satellite navigation for fuel efficiency and space debris avoidance
- ❖ Focused on trajectory prediction and collision avoidance to support sustainable orbital operations

Organizations

Software Department Head

Autonomous Maritime Robotics Association **2022–Present**

- ❖ Built a supportive, growth-oriented team culture while improving retention and training efficiency
- ❖ Led the development of autonomous underwater vehicle (AUV) software integrating CV, sonar data, and ML-based decision-making
- ❖ Guiding undergrads in publishing research

Achievements & Awards

IEEE-Eta Kappa Nu Best Student Paper Award **2025**

- ❖ Awarded IEEE-Eta Kappa Nu Best Paper Award, selected from 280 papers

DOD SMART (Science, Mathematics, and Research for Transformation) Scholarship **2023–Present**

- ❖ A competitive program funded by the Department of Defense
- ❖ Recognizes and supports exceptional academic achievements and commitment to pursuing careers in STEM fields

1st Place In Embry-Riddle Student Research Symposium **2023**

- ❖ Awarded after competing against 78 other researchers

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- ❖ Platform to showcase research scholarly pursuits through in-person presentations

Publications

Primary Author

Dueling Deep Q-Learning for Intrusion Detection

2024–2025

Objective: To design an adaptive and interpretable reinforcement learning framework for network intrusion detection, leveraging dueling Q-learning to enhance decision stability, classification accuracy, and explainability across diverse cyberattack types.

- ❖ *Conference: Accepted to IEEE SoutheastCon 2025*
- ❖ Awarded IEEE-Eta Kappa Nu Best Paper Award, selected from 280 papers
- ❖ Developed a dueling Deep Q-Network (DQN) in PyTorch for multiclass intrusion detection, separating state-value and action-advantage streams for enhanced stability and convergence.
- ❖ Trained on CIC-IDS2018 (2.18M samples across DDoS, DoS, Botnet, Brute-force), achieving 99.68% weighted accuracy, outperforming prior RL-based IDS by >11%.
- ❖ Designed a custom OpenAI Gym environment to simulate sequential network traffic and reward-based actions, enabling adaptive detection across evolving threat landscapes.
- ❖ Integrated SHAP (SHapley Additive Explanations) for interpretability, identifying key predictive features (e.g., RST Flag Count, Flow Pkts/s) and increasing operational transparency.
- ❖ Future extensions include hierarchical attack-type classification and deployment in enterprise-scale network simulations

Primary Author

Signal-Centric Remote Sensing via Alternative Preprocessing and Acoustic Processing for ML-Driven Applications

2023–2025

Objective: To develop a computationally efficient sonar data processing pipeline that transforms raw acoustic signals into machine learning-ready representations, achieving robust denoising, noise resilience, and accurate object detection in underwater environments.

- ❖ *Conference: Accepted to IEEE SoutheastCon 2025*
- ❖ Introduced a CSV-based sonar data pipeline, reducing total processing time by 91.18% compared to image-based approaches while improving object detection accuracy.
- ❖ Designed machine learning preprocessing using median filtering and background subtraction to enhance SNR and PSNR, improving denoised sonar scan clarity.
- ❖ Implemented K-Means clustering with convex hull estimation for underwater object detection, validated on BlueRobotics Ping360 sonar aboard the Hammerhead AUV.
- ❖ Demonstrated robust detection of buoys and gates with faster runtime and higher fidelity than image-based methods.

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Sole Author

Milo: A Magnitude-Invariant Learning Optimizer with Group-Wise Gradient Normalization **2025**

Objective: To formulate a magnitude-invariant first-order optimization algorithm, built upon Stochastic Gradient Descent (SGD), that decouples update scaling from gradient magnitude through group-wise normalization, achieving stable convergence, robust generalization, and competitive accuracy with state-of-the-art adaptive methods.

- ❖ *Journal: Submitted to Transactions on Machine Learning Research (TMLR), under review*
- ❖ Proposed the Magnitude Invariant Learning Optimizer(Milo), a first-order optimizer extending SGD with grouped gradient normalization and scale-aware mixing, enabling magnitude-invariant updates without second-moment tracking.
- ❖ Validated across seven benchmarks, including loss landscape navigation, logistic regression, MLPs, CNNs (ResNet34, VGG11), and 3D reinforcement learning environments.
- ❖ Demonstrated competitive performance across diverse tasks, rivaling and often surpassing adaptive methods like AdamW, AdaGrad, and NovoGrad, while maintaining first-order simplicity and runtime efficiency.
- ❖ Empirical results showed improvements in FL and loss scores when benchmarked with AdamW, SGD, AdaGrad, NovoGrad, and other popular optimizers.
- ❖ Achieved comparable runtime despite normalization overhead, validating efficiency and scalability.
- ❖ Proved sublinear convergence ($O(1/\sqrt{T})$) in convex and non-convex settings, with improved stability and flatter loss landscapes.
- ❖ Future work may include exploring integration with second-moment estimation to combine Milo's scale-invariant grouping with adaptive variance tracking, potentially bridging first- and second-order advantages.

Primary Author

Satellite Trajectory Optimization via Machine Learning for Space Debris Avoidance **2024–2025**

Objective: To create an autonomous reinforcement learning framework for fuel-efficient satellite collision avoidance, optimizing orbital trajectories in high-fidelity astrodynamics simulations to support sustainable space operations and debris mitigation.

- ❖ *Journal: Submitted to IEEE Access, under review*
- ❖ Designed a reinforcement learning (RL) framework using Proximal Policy Optimization (PPO) in a custom astrodynamics simulator modeling Sun/Moon perturbations and fuel-aware thrust.
- ❖ Achieved 97.5% collision avoidance success across 1,000 deterministic GEO episodes, outperforming rule-based (20.7%) and impulsive Δv (27.5%) baselines.
- ❖ Built a Gym-compatible open-source environment with Newtonian dynamics, deterministic replay, and telemetry logging for reproducible evaluation.
- ❖ Used curriculum learning and reward shaping to balance collision avoidance, fuel conservation, and orbital stability.
- ❖ Supports autonomous space traffic management and sustainable orbital operations through intelligent, reproducible collision avoidance.

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Projects

Project Member

Reinforcement Learning for Aerial Robotics: Sim-to-Real Transfer

2024–Present

Objective: To develop software for the identification and localization of aircraft, aiming to operate primarily independent of Federal Aviation Administration systems to promote enhanced aerial safety

- ❖ Ongoing Research – Flexible & Intelligent Complex Systems Research Group
- ❖ Developing a reinforcement learning (RL) control policy for Crazyflie 2.X nano-quadrotors, trained in NVIDIA Omniverse Isaac Gym for a scalable physics-based simulation.
- ❖ Focused on bridging the Sim-to-Real gap by transferring learned PPO/SAC policies from simulation to real drones, evaluating differences in stability, drift, and trajectory tracking.
- ❖ Designed training curriculum for hover stabilization and figure-eight trajectory tracking under noise, wind, and mass variation perturbations.
- ❖ Producing a quantitative Sim-to-Real analysis comparing simulation and hardware metrics, with structured reporting on gap sources and mitigation strategies.

Project Lead

Enhancing Underwater Object Recognition with Computer Vision and Filtering Techniques

2023–Present

Objective: To implement multiple Computer Vision models to increase object recognition efficiency in underwater environments

- ❖ Utilized Yolo v8 (Object detection & classification), SuperPoint (feature point detection software), and SuperGlue (image matching software) for live data analysis from an autonomous underwater vehicle.
- ❖ Formulated an algorithm to utilize Kalman Filter predictions in combination with YoloV8 for dynamic and continuous object location tracking.
- ❖ Planned prediction of confidence for object location tracking and classification based off multiple outputs of detection from each Computer Vision model.

References

Dr. Alex Vargas Acosta

Embry-Riddle Aeronautical University

Relationship: Research Advisor

Title: Assistant Professor

Email: vargasar@erau.edu

Eric Montag

Naval Surface Warfare Center Dahlgren Division

Relationship: Group Lead and Mentor

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Embry-Riddle Aeronautical University

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